Camera-Only Multi-View BEV Occupancy Estimation Under Diverse Weather Conditions

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I. Abstract

Bird's-Eye View (BEV) perception enables compact top-down scene understanding but remains challenging for camera-only systems, particularly under adverse weather. We study multi-camera BEV occupancy estimation in clear, rainy, and foggy conditions using a synthetic CARLA dataset with six RGB cameras. BEV pseudo-labels are generated using a weakly supervised heuristic that maps YOLOv8 image-space detections into a top-down BEV grid. A lightweight BEVUNet-style encoder–decoder is trained to predict binary BEV occupancy maps. Results demonstrate strong performance in clear conditions and graceful degradation under rain and fog, with weather-specific inference thresholds improving precision without sacrificing recall.

Code is available at https://github.com/Avantikarattan6134/CARLA-PerceptionLab.

II. Introduction

Reliable perception is fundamental to autonomous driving, enabling tasks such as planning and control. BEV representations have become central due to their geometric consistency and suitability for spatial reasoning [2], [3]. Most high-performing BEV models fuse LiDAR, radar, and cameras [4], but such systems depend on expensive sensors and extensive annotations. Camera-only BEV perception is more scalable but lacks explicit depth cues and requires robust multi-view feature aggregation [5].

We study camera-only BEV occupancy estimation under adverse weather conditions using a synthetic CARLA dataset with six RGB cameras providing full 360° coverage. Experiments are conducted in clear, rainy, and foggy settings to systematically evaluate robustness to visibility degradation. To avoid manual BEV annotation, we adopt a weakly supervised pseudo-labeling strategy that maps YOLOv8 image-space detections into a top-down BEV grid using simple perspective cues rather than explicit geometric projection, following recent work on weakly supervised BEV learning [7], [8].

Our method uses a BEVUNet-style encoder–decoder trained with Adam and Binary Cross-Entropy loss to predict binary occupancy maps. We additionally introduce weather-specific confidence thresholds at inference to improve precision under rain and fog. Experiments demonstrate that lightweight BEV convolutional models can achieve competitive performance without transformers or 3D backbones, highlighting a practical path toward scalable camera-only BEV perception.

III. Related Work

A. Camera-Based BEV Perception

Camera-only BEV perception has progressed rapidly in recent years. Early methods such as Lift-Splat-Shoot (LSS) [2] rely on estimating depth to lift image features into BEV space, inspiring many frustum and voxel-based formulations. Transformer-based approaches form a second paradigm, BEVFormer [3] and PETR [5] leveraging attention mechanisms to aggregate multi-view features into BEV representations. Other works, including M^2BEV [11], SurroundOcc [12], and OccFormer [13], focus on dense BEV segmentation and occupancy prediction. While effective, these approaches require substantial supervision with high computational cost.

B. U-Net and BEV-Space CNNs

U-Net [15] is a widely adopted encoder–decoder architecture for dense prediction, whose skip connections enable accurate spatial reconstruction. BEVNet-style architectures [14] extend image segmentation by applying 2D CNNs directly in BEV space after projecting image features onto the ground plane, often via inverse perspective mapping or explicit geometric transformations. Such methods perform well in structured environments but can be sensitive to calibration errors, depth ambiguities, and adverse weather conditions.

C. Proposed BEVUNet Architecture

We propose *BEVUNet*, a lightweight convolutional architecture designed for camera-only BEV occupancy estimation under weak supervision and adverse weather. Inspired by BEV-space CNNs and U-Net-style encoder–decoder designs [14], [15], BEVUNet processes multi-view RGB inputs using shared-weight encoders and fuses features across cameras before decoding directly into a binary BEV occupancy grid.

BEVUNet avoids depth supervision and 3D voxelization, resulting in a simpler and more efficient architecture. This design is well suited for learning from noisy pseudo-labels and enables scalable camera-only BEV perception under resource constraints and degraded visibility.

IV. Dataset and Pseudo-Label Generation

A. Synthetic Multi-View Dataset

We use a synthetic multi-view RGB dataset generated with the CARLA simulator, consisting of six synchronized cameras mounted around the ego vehicle to provide full 360° coverage, with denser sampling in the forward direction. The dataset includes clear, rainy, and foggy weather conditions, enabling controlled evaluation under increasing visual degradation.

Each camera image is resized to 128×128 pixels for computational efficiency. Although CARLA provides accurate intrinsic and extrinsic calibration parameters (listed in Appendix A), they are not used directly in our pseudo-labeling pipeline; instead, we rely on a heuristic image-to-BEV mapping described in the following subsection.

(a) Foggy (b) Rainy (c) Clear

Fig. 1: Example RGB images from CARLA illustrating progressive visual degradation across weather conditions.

B. Pseudo-Label Generation Using Object Detection

To avoid costly manual BEV annotation, we employ a pseudo-labeling strategy based on image-space object detection. A pretrained YOLOv8 model is applied independently to each camera view to detect traffic-relevant objects. YOLOv8 follows a backbone–neck–head architecture, enabling efficient multi-scale detection as shown in Figure 2.

Detections are produced in normalized image coordinates using the standard YOLO format. Weather-specific confidence thresholds are applied to account for detection reliability under different visual conditions, using higher thresholds in clear scenes and lower thresholds in foggy and rainy environments.

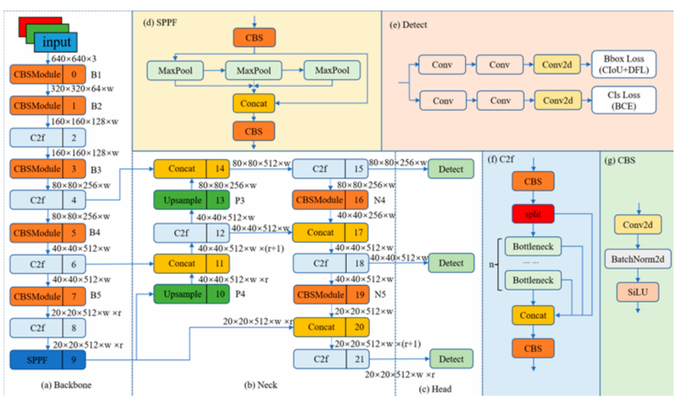

Fig. 2: Architecture of YOLOv8 consisting of a backbone, neck, and detection head. The model is used for image-space object detection to generate BEV pseudo-labels. Figure adapted from [9].

C. Heuristic Projection from Image Space to BEV

Instead of explicit geometric projection using camera intrinsics and extrinsics, we employ a lightweight heuristic mapping to convert image-space detections into BEV pseudo-labels. This approach avoids per-frame 3D reconstruction while providing consistent top-down supervision suitable for weakly supervised BEV learning.

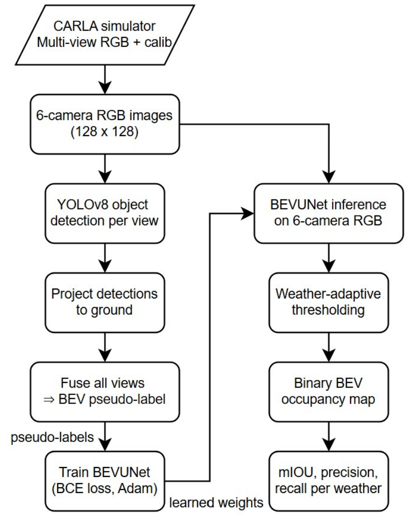

Fig. 3: Overview of the proposed pipeline.

We define a 128×128 BEV grid covering a $60\,\mathrm{m} \times 60\,\mathrm{m}$ region in front of the ego vehicle, with lateral extent $X \in [-30, 30]$ m and longitudinal extent $Y \in [0, 60]$ m. Each YOLOv8 detection is represented by its normalized bounding-box center $(c_x^{(n)}, c_y^{(n)}) \in [0, 1]^2$.

Using simple perspective priors, the horizontal image coordinate maps to BEV lateral position and the vertical coordinate acts as a monotonic proxy for depth:

$$X = (c_x^{(n)} - 0.5)(X_{\max} - X_{\min}), \quad Y = Y_{\min} + (1 - c_y^{(n)})(Y_{\max} - Y_{\min}),$$

with $X_{\min} = -30$, $X_{\max} = 30$, $Y_{\min} = 0$, and $Y_{\max} = 60$.

Each projected (X, Y) location is discretized to the nearest BEV cell, and a small neighborhood around the cell is marked occupied to account for localization noise. This procedure is applied independently to all six cameras, and the resulting BEV maps are fused by union across views. While not metrically accurate, this heuristic mapping yields consistent BEV pseudo-labels sufficient for training a camera-only BEV occupancy network under weak supervision.

D. Dataset Summary

The final dataset consists of synchronized multi-view RGB images, corresponding with BEV pseudo-labels, and camera calibration parameters for each frame. This setup supports systematic evaluation of camera-only BEV occupancy estimation under weak supervision and varying weather conditions.

The intrinsic (K) and extrinsic matrices (T) are shown in Appendix A.

V. METHODOLOGY

This section describes the problem formulation, network architecture, training objective, and inference strategy for camera-only BEV occupancy estimation. As shown in Fig. 3, the pipeline consists of pseudo-label generation using YOLOv8, BEVUNet training, and weather-adaptive inference.

A. Problem Formulation

Given six synchronized RGB images $\{I_i\}_{i=1}^{6}$, the goal is to predict a binary Bird's-Eye View (BEV) occupancy map indicating ground-plane locations of traffic-relevant objects. The task is formulated as

$$B = f(I_1, I_2, \ldots, I_6),$$

where $B \in \{0,1\}^{128\times 128}$ represents a BEV grid covering a $60\,\mathrm{m} \times 60\,\mathrm{m}$ region in front of the ego vehicle, with lateral extent $X \in [-30, 30]$ m and longitudinal extent $Y \in [0, 60]$ m. Learning is performed under weak supervision using pseudo-labels.

B. Network Architecture

We employ a U-Net–style encoder–decoder architecture, termed *BEVUNet*, for camera-only BEV prediction. Each camera image is processed by a shared-weight encoder, and multi-view features are fused into a unified representation. A decoder with skip connections produces a single-channel BEV logits map. Spatial reasoning is learned implicitly from multi-view supervision without explicit geometric constraints.

C. Training Objective

BEVUNet is trained using pseudo-labeled BEV occupancy maps derived from object detections. Training is formulated as pixel-wise binary classification using Binary Cross-Entropy loss with logits:

$$\mathcal{L} = \mathrm{BCEWithLogitsLoss}(B_{\mathrm{pred}}, B_{\mathrm{gt}}),$$

where B_{pred} denotes predicted logits and B_{gt} denotes pseudo-ground-truth labels. The model is optimized end-to-end using the Adam optimizer.

D. Inference and Weather-Adaptive Thresholding

During inference, the predicted logits are converted to probabilities via a sigmoid function and binarized using weather-specific thresholds τ_w:

$$\hat{B}(x,y) = \begin{cases} 1, & \sigma(B_{\mathrm{pred}}(x,y)) \geq \tau_w, \\ 0, & \text{otherwise.} \end{cases}$$

Higher thresholds are used in clear conditions to suppress false positives, while lower thresholds are applied in rainy and foggy scenes to preserve recall, thus improving robustness under degraded visibility.

VI. Experimental Setup

All experiments are conducted on a synthetic multi-view dataset generated using the CARLA simulator. The dataset is split into training and test sets using a 4:1 ratio on a per-scene basis to avoid temporal leakage, with each split containing samples from clear, rainy, and foggy weather conditions. Each sample consists of six synchronized RGB images resized to 128×128 pixels and a corresponding BEV pseudo-label represented on a 128×128 grid covering a $60\,\mathrm{m} \times 60\,\mathrm{m}$ region in front of the ego vehicle.

BEVUNet is trained end-to-end on pseudo-labeled BEV occupancy maps using the Adam optimizer with a learning rate of 1×10^{-4} and Binary Cross-Entropy loss with logits. Training is performed for up to 10 epochs and the checkpoint achieving the lowest validation loss is used for evaluation. No explicit data augmentation or geometric regularization is applied, allowing the model to learn robust multi-view feature aggregation directly from noisy pseudo-labels.

During inference, predicted BEV logits are converted to probabilities using a sigmoid activation and binarized with weather-specific confidence thresholds: $\tau = 0.55$ for clear, $\tau = 0.45$ for rainy, and $\tau = 0.40$ for foggy conditions. This adaptive thresholding improves precision in visually clean scenes while preserving recall under degraded visibility.

Model performance is evaluated using Mean Intersection-over-Union (mIoU), precision, recall, and Mean Average Precision (mAP) computed on the BEV occupancy maps. Metrics are averaged across test samples and reported separately for each weather condition to assess robustness. All experiments are implemented in PyTorch and executed on a single GPU. YOLOv8 is used exclusively for pseudo-label generation and is not fine-tuned during BEVUNet training.

VII. Results and Analysis

This section presents quantitative and qualitative evaluations of the proposed camera-only BEV occupancy framework, with emphasis on robustness across different weather conditions.

A. Quantitative Results

BEVUNet is evaluated using mIoU, precision, recall, and mAP under clear, rainy, and foggy conditions, with weather-adaptive thresholds applied during inference. As shown in Fig. 4 in Appendix B, both training and validation losses decrease steadily across epochs with closely aligned trends, indicating stable optimization and good generalization despite weak supervision.

TABLE I: Per-weather BEV occupancy prediction performance.

Weather	mIoU	Precision	Recall	mAP
Clear	0.341	0.342	0.979	0.846
Rainy	0.263	0.265	0.981	0.706
Foggy	0.250	0.252	0.953	0.727

As shown in Table I, BEVUNet achieves reliable BEV occupancy prediction across diverse weather conditions, with the highest performance observed in clear weather (mIoU = 0.341) and consistently high recall above 0.95 in all scenarios. Although mIoU and precision decrease under rainy and foggy conditions due to degraded visibility, the model maintains competitive mAP values, demonstrating robust multi-view BEV learning under adverse weather.

B. Comparison with Existing BEV Methods

Table II compares BEVUNet with representative camera-based BEV methods under clear and adverse weather conditions. In clear weather, BEVUNet achieves higher mAP

TABLE II: mAP and mIoU comparison with related BEV-based methods.

Method	Weather	mAP	mIoU
PETR [5]	Clear	0.370	-
PETRv2 [6]	Clear	0.421	-
BEVFormer [3]	Clear	0.416	0.487
BEVDet [4]	Clear	0.397	-
BEVDet4D [4]	Clear	0.396	-
DA-BEV [10]	Clear	0.203	0.356
DA-BEV [10]	Rainy	0.110	-
Ours (BEVUNet)	Clear	0.846	0.341
Ours (BEVUNet)	Rainy	0.706	0.263
Ours (BEVUNet)	Foggy	0.727	0.250

than prior CNN-based approaches such as BEVDet and DA-BEV, and remains comparable to transformer-based methods including BEVFormer and PETRv2 despite its lighter architecture. Under rainy and foggy conditions, BEVUNet maintains strong mAP performance, indicating improved robustness to visibility degradation relative to methods whose performance drops significantly under adverse weather.

C. Effect of Weather-Adaptive Thresholding

Weather-specific thresholding consistently improves precision while preserving recall, particularly in rainy and foggy conditions. This demonstrates that simple post-processing can effectively reduce false positives under varying visibility.

D. Qualitative Analysis

Qualitative results align with quantitative trends. Clear scenes yield compact and spatially consistent occupancy maps, while rain and fog introduce increasing noise due to weakened depth cues. Despite this, major occupied regions are consistently recovered across all conditions.

E. Summary of Results

Overall, the results demonstrate that camera-only BEV occupancy estimation is feasible across diverse weather conditions by using multi-view aggregation and adaptive thresholding. While heuristic pseudo-labels limit metric accuracy, the model achieves stable coverage and competitive performance relative to more complex BEV methods.

VIII. Discussion

This study demonstrates that camera-only BEV occupancy estimation can be learned effectively under weak supervision without LiDAR. Multi-view feature aggregation and weather-adaptive thresholding enable robust performance across clear, rainy, and foggy conditions. Performance degradation in adverse weather primarily stems from the limitations of RGB-only perception and noise in pseudo-labels. Nevertheless, BEVUNet achieves competitive results relative to more complex BEV methods, highlighting the viability of lightweight architectures for BEV perception.

A key limitation of the proposed approach is that BEV pseudo-labels are generated using a heuristic image-to-BEV mapping rather than exact geometric projection. Consequently, the resulting occupancy maps are non-metric and provide only approximate spatial localization. Despite this, the mapping remains consistent across views and frames, enabling the network to learn robust spatial patterns from noisy supervision.

TABLE III: Summary of BEVUNet architecture for a 128×128 BEV grid.

Stage	Operation	Channels	Resolution
Input	Multi-view RGB images	18	128×128
Enc 1	$2\times$ Conv–BN–ReLU	32	128×128
Enc 2	MaxPool, $2\times$ Conv–BN–ReLU	64	64×64
Enc 3	MaxPool, $2\times$ Conv–BN–ReLU	128	32×32
Dec 2	Transposed Conv, concat Enc 2, $2\times$ Conv	64	64×64
Dec 1	Transposed Conv, concat Enc 1, $2\times$ Conv	32	128×128
Output	1×1 Conv (logits)	1	128×128

A. Comparison with Existing BEV Architectures

Many BEV perception methods rely on explicit geometry, depth estimation, or transformer-based attention to construct BEV representations from multi-camera inputs [2], [3], [4], often requiring strong supervision and high computational cost.

In contrast, BEVUNet adopts a lightweight convolutional design and learns BEV occupancy implicitly through multi-view feature aggregation under weak supervision. Training relies on heuristic pseudo-labels derived from 2D object detections rather than LiDAR-based ground truth or manual BEV annotations [4], [8], enabling a scalable and cost-effective pipeline.

By focusing on binary BEV occupancy instead of full 3D detection or semantic segmentation, BEVUNet further reduces architectural complexity while remaining well suited for downstream planning and collision avoidance, offering a practical camera-only solution with improved robustness to adverse weather.

B. BEVUNet Architecture

BEVUNet employs a lightweight U-Net–style encoder–decoder with shared weights across six camera views and explicit multi-view feature fusion. Each RGB image is processed by a shared encoder that extracts a three-level feature hierarchy with progressively decreasing spatial resolution.

Multi-view features are fused by averaging at each pyramid level, producing a unified representation that is decoded using transposed convolutions and skip connections. A final 1×1 convolution outputs a single-channel BEV logits map, supervised using `BCEWithLogitsLoss`. During inference, a sigmoid activation followed by weather-specific thresholding produces the binary BEV occupancy prediction.

Table III summarizes the main stages of the BEVUNet architecture for a 128×128 BEV grid.

IX. Future Work

Future improvements include enhancing pseudo-label quality via stronger detectors or temporal filtering, incorporating depth-aware or monocular depth features to strengthen geometric reasoning, and applying temporal fusion to improve consistency and reduce false positives. Extending the system to multi-class BEV prediction and evaluating on real-world datasets will further broaden its applicability.

X. References

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Appendix A: Intrinsic (K) and Extrinsic (T) Matrices

The following camera intrinsic and extrinsic matrices are provided by CARLA and are included for completeness and potential future extensions. The current pseudo-labeling pipeline does not use these parameters directly, instead relying on a heuristic image-to-BEV mapping as described in Section IV-C.

$$K = \begin{bmatrix} 640.0 & 0.0 & 640.0 \\ 0.0 & 640.0 & 360.0 \\ 0.0 & 0.0 & 1.0 \end{bmatrix}$$

$$T_{front} = \begin{bmatrix} 0.9999961 & -0.0027785 & 0.0 & -63.0448 \\ 0.0027785 & 0.9999961 & 0.0 & 24.4755 \\ 0.0 & 0.0 & 1.0 & 1.5018 \\ 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix}$$

$$T_{back} = \begin{bmatrix} -0.9999961 & 0.00277867 & 0.0 & -66.144836 \\ -0.0027786 & -0.9999961 & 0.0 & 24.4668426 \\ 0.0 & 0.0 & 1.0 & 1.50183939 \\ 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix}$$

$$T_{right} = \begin{bmatrix} -0.0027783 & -0.9999961 & 0.0 & -64.646507 \\ 0.99999612 & -0.0027783 & 0.0 & 25.0710086 \\ 0.0 & 0.0 & 1.0 & 1.50183939 \\ 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix}$$

$$T_{left} = \begin{bmatrix} 0.0027786 & 0.9999961 & 0.0 & -64.64318 \\ -0.999996 & 0.0027786 & 0.0 & 23.871011 \\ 0.0 & 0.0 & 1.0 & 1.5018393 \\ 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix}$$

$$T_{front-right} = \begin{bmatrix} 0.7051392 & -0.709068 & 0.0 & -63.24596 \\ 0.7090688 & 0.7051392 & 0.0 & 24.874898 \\ 0.0 & 0.0 & 1.0 & 1.5018393 \\ 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix}$$

$$T_{front-left} = \begin{bmatrix} 0.7090689 & 0.7051391 & 0.0 & -63.24373 \\ -0.705139 & 0.7090689 & 0.0 & 24.074901 \\ 0.0 & 0.0 & 1.0 & 1.5018393 \\ 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix}$$

Appendix B: Additional Qualitative Results

This appendix contains additional qualitative BEV occupancy prediction results across multiple weather conditions. Figure 4 represents the graph between validation and training loss. Figures 5, 6 and 7 illustrate representative samples from clear, rainy, and foggy scenes respectively.

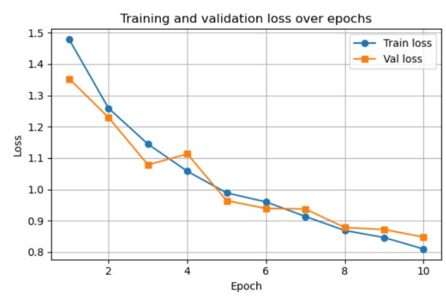

Fig. 4: Training and validation loss for BEVUNet.

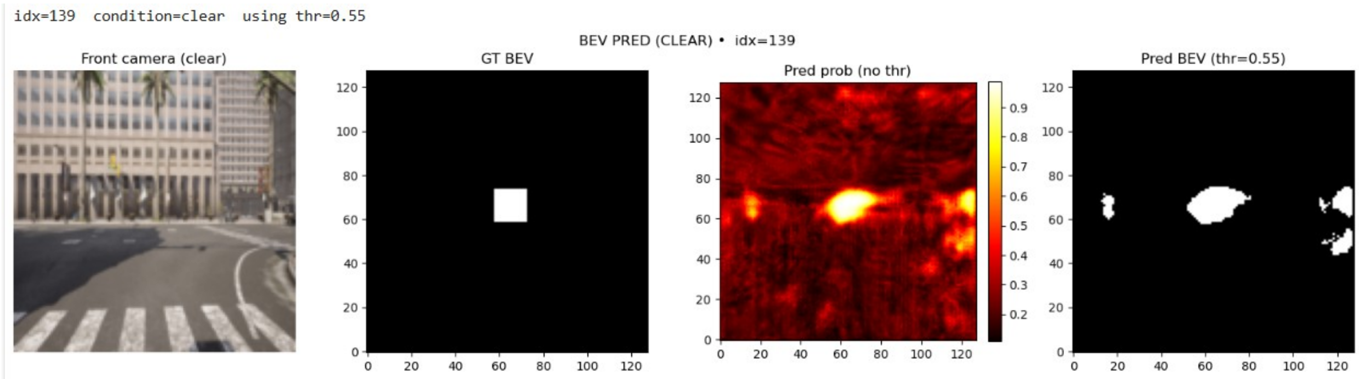

Fig. 5: Qualitative BEV occupancy prediction under clear weather conditions. From left to right: front camera RGB image, pseudo ground-truth BEV, predicted BEV probability map, and final binary BEV prediction after thresholding ($\tau = 0.55$).

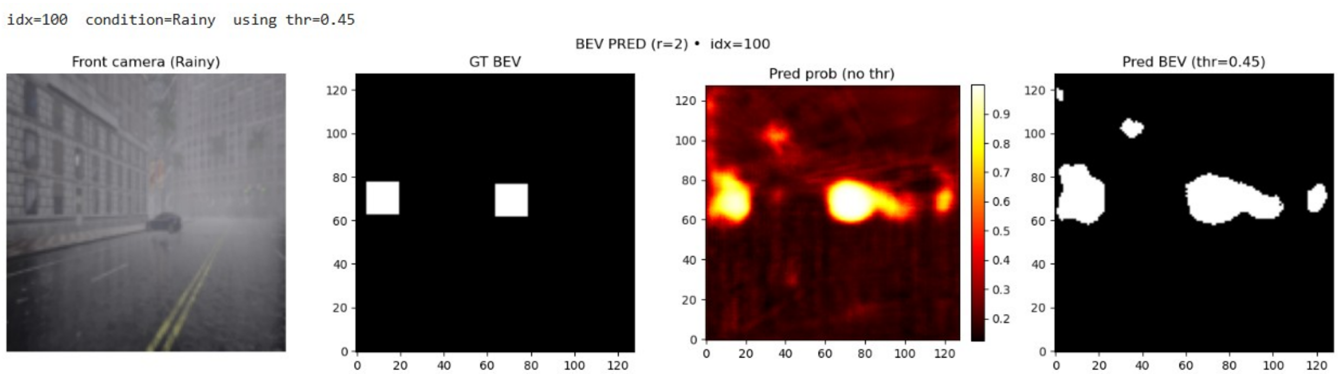

Fig. 6: Qualitative BEV occupancy prediction under rainy weather conditions. From left to right: front camera RGB image, pseudo ground-truth BEV, predicted BEV probability map, and final binary BEV prediction after thresholding ($\tau = 0.45$).

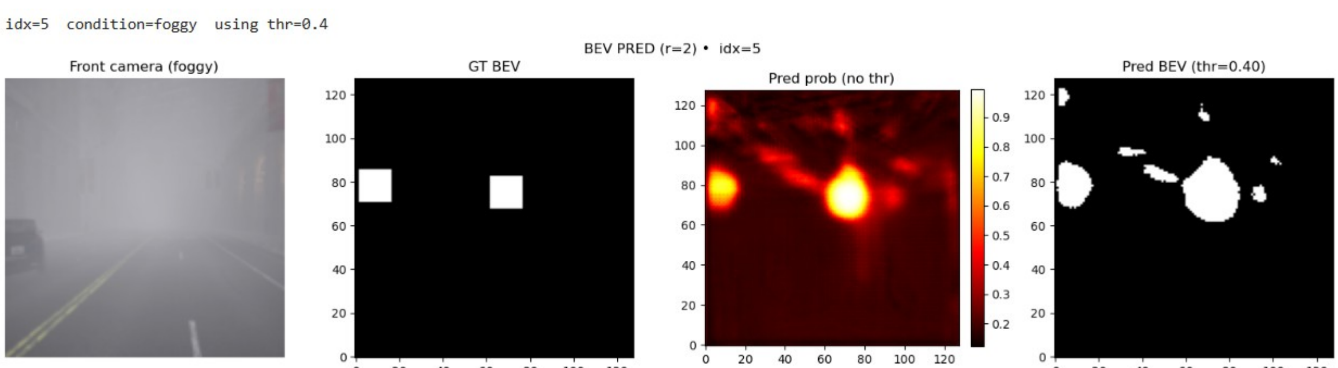

Fig. 7: Qualitative BEV occupancy prediction under foggy weather conditions. From left to right: front camera RGB image, pseudo ground-truth BEV, predicted BEV probability map, and final binary BEV prediction after thresholding ($\tau = 0.40$).